Research Note 001

The Invisible Patient

Genomic Representation, AI Governance, and Breast Cancer Prediction in East Africa

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Executive Summary

Artificial intelligence is reshaping cancer care through diagnostic imaging, genomic risk prediction, and precision medicine. Yet these systems are only as equitable as the data used to build them. Across biomedical AI, African populations remain critically underrepresented in training datasets, raising concerns about the fairness, reliability, and clinical relevance of technologies increasingly promoted as solutions to Africa's cancer burden.

This research note argues that the absence of East African women from biomedical datasets is not merely a technical oversight but a social and political condition requiring social and political analysis. Questions of algorithmic fairness cannot be separated from questions of representation, governance, and scientific sovereignty.

Using triple-negative breast cancer (TNBC) as a case study, this note explores how genomic underrepresentation may shape AI-driven cancer prediction and risk assessment in East Africa. It proposes that equitable AI development requires more than improved model performance; it requires attention to the structural conditions that determine whose data are collected, whose biology becomes the reference standard, and whose health outcomes are prioritized.

Conceptual Framework

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Underrepresentation in Genomic Research
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Underrepresentation in Biomedical Datasets
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AI Models Trained on Incomplete Populations
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Reduced Reliability for East African Patients
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Diagnostic and Prediction Disparities
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Health Inequities
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Need for Data Justice, AI Governance,
and Scientific Sovereignty
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This framework situates algorithmic inequity not as a failure of technology alone, but as the downstream consequence of historical, institutional, and political decisions about data production and scientific participation.

The Problem

Breast cancer remains one of the leading causes of cancer mortality among women globally. In East Africa, the burden is intensified by late-stage diagnosis, limited access to specialized oncology services, and unequal access to emerging precision medicine technologies.

At the same time, AI systems are increasingly being developed to support cancer risk assessment, diagnosis, and treatment decisions. Their effectiveness, however, depends on the populations represented within the datasets used to train and validate them.

Africa contains more than 70% of global human genetic diversity yet accounts for approximately 2% of participants in genome-wide association studies. Only 1% of global health data originates from African countries. A 2025 scientometric analysis of 264 AI mammography studies found no datasets from low-income countries and no first authors affiliated with low-income country institutions. Sub-Saharan Africa was effectively absent from the research landscape.

When populations are missing from training data, the consequences extend beyond exclusion. AI systems may systematically underperform for those populations, contributing to delayed diagnosis, inaccurate risk assessment, and unequal access to the benefits of precision medicine.

Why East Africa and Why TNBC?

East African populations remain among the most genomically underrepresented groups in global biomedical research while simultaneously becoming targets of AI-enabled healthcare innovation. This imbalance reflects broader structural inequalities in research funding, data infrastructure, and global knowledge production.

Triple-negative breast cancer provides a particularly important lens through which to examine these issues. TNBC is the most aggressive breast cancer subtype, characterized by limited targeted treatment options and poorer survival outcomes. Evidence suggests that TNBC occurs at substantially higher rates in several East African populations than in European populations, while emerging genomic studies reveal distinct biological characteristics that are rarely reflected in the datasets used to develop contemporary AI systems.

Recent work demonstrates that polygenic risk scores developed specifically for women of African ancestry outperform European-derived models, yet substantial performance gaps remain. Similarly, AI-based histopathology systems trained primarily on high-income-country datasets have shown disparities in performance among Black patient populations.

The challenge is therefore not whether AI can improve cancer care in Africa. The challenge is whether AI systems are being developed from data that adequately represent the populations they are intended to serve.

Research Questions

This research programme is organized around four interconnected questions:

Structural: What historical, institutional, and political-economic factors have contributed to the underrepresentation of East African women in genomic and clinical AI datasets?

Technical-Social: How does dataset imbalance influence the performance and fairness of AI-based breast cancer prediction systems when applied to East African populations?

Governance: How do existing health data governance frameworks in East Africa enable or constrain equitable AI development?

Normative: What alternative governance approaches, including Ubuntu-informed and data justice frameworks, can support more accountable and context-sensitive biomedical AI?

What the Literature Tells Us — and What It Does Not

The literature clearly establishes three realities.

First, African populations remain substantially underrepresented in genomic and biomedical datasets despite their extraordinary genetic diversity.

Second, the biological characteristics of breast cancer, particularly TNBC, differ across populations in ways that have important implications for precision medicine and AI-driven prediction systems.

Third, concerns about algorithmic bias, data extraction, and unequal participation in AI development have become central themes within global debates on responsible AI.

Yet important gaps remain. The intersection of East African genomics, TNBC biology, biomedical AI, and health data governance remains largely unexplored. Existing governance studies rarely address oncological AI specifically, while Ubuntu-informed approaches to AI governance have yet to be systematically applied to cancer prediction systems in East Africa.

These gaps are not peripheral. They are precisely the points at which questions of representation, trust, and health equity become most consequential.

Why This Matters

Discussions about AI in healthcare frequently emphasize model accuracy. Accuracy alone, however, does not guarantee fairness.

The benefits of AI cannot be assumed to extend equally across populations when the underlying data fail to reflect human diversity. Underrepresentation is not simply a data problem; it is a governance problem, a research infrastructure problem, and ultimately a justice problem.

Addressing these challenges requires moving beyond technical optimization toward broader questions of participation, accountability, and scientific sovereignty. If AI is to contribute meaningfully to cancer care in East Africa, the communities most affected by these technologies must also be represented in the data, institutions, and governance structures that shape them.

The future of equitable biomedical AI depends not only on better algorithms, but on whose knowledge, biology, and experiences are included in their design.

About the Author

Joy Manyasi Kabaka is a molecular biologist and founder of Sequence Africa, a platform documenting African genomics, biodiversity, and emerging technologies. Her work explores the intersections of genomics, artificial intelligence, scientific sovereignty, and science communication. She is a member of the Genomics and Biotechnology Working Group at the Africa Bioethics Network.

This research note forms part of an ongoing research programme examining algorithmic inequity, genomic representation, and health data justice in East Africa.